

Bifurcations

$$\dot{x} = f(x, \lambda), \quad x \in \mathbb{R}^n, \quad \lambda \in \mathbb{R}$$

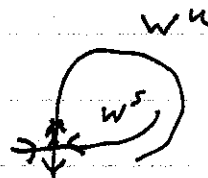
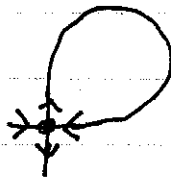
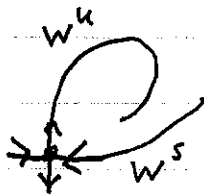
"Local bifurcations" : can be identified via a linear stability calculation

steady state $(n \geq 1)$

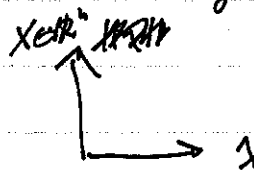
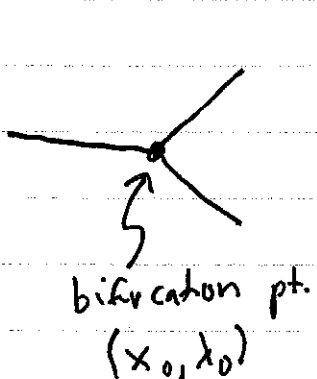
Hopf $(n \geq 2)$

"Global bifurcations" : require information outside of a small neighborhood of an equilibrium

example: homoclinic bifurcation



Steady state: What is a necessary condition for the change in # of fixed-pts of a vector field as we change the parameter λ ?



$f(x, \lambda) = 0$ is eqn. for steady states

Say we know a soln. (x_0, λ_0) and we want to determine how x_0 moves with λ , meaning to uniquely "continue" it in λ :



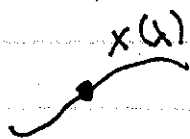
$$f(x_0 + \delta x, \lambda_0 + \epsilon) = f(x_0, \lambda_0) + D_x f(x_0, \lambda_0) \delta x + \frac{\partial f}{\partial \lambda}(x_0, \lambda_0) \epsilon + \dots = 0$$

$$\delta x \approx -[D_x f(x_0, \lambda_0)]^{-1} \frac{\partial f}{\partial \lambda}(x_0, \lambda_0) \epsilon$$

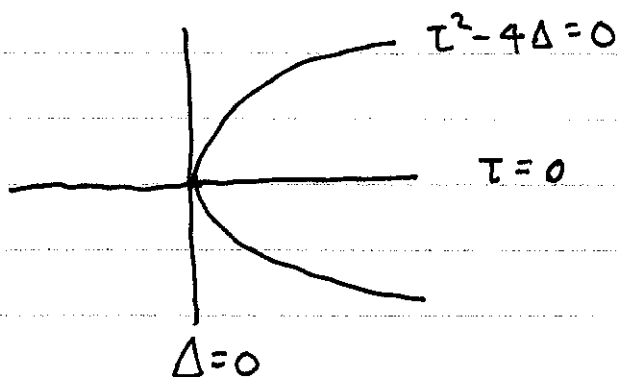
assumes $D_x f(x_0, \lambda_0)$ is invertible, which is true if $\text{Det}[D_x f(x_0, \lambda_0)] \neq 0$

$$\text{Steady state bifurcation: } \begin{cases} f(x_0, \lambda_0) = 0 \\ \text{Det}[D_x f(x_0, \lambda_0)] = 0 \end{cases}$$

Implicit function thm: Suppose $f: \mathbb{R}^n \times \mathbb{R} \rightarrow \mathbb{R}^n$ is a continuously differentiable function of $(x_1, \dots, x_n) \in \mathbb{R}^n$ and $\lambda \in \mathbb{R}$, and $f(0, 0) = 0$. If Jacobian matrix $Df(0, 0)$, where $Df_{ij} = \frac{\partial f_i}{\partial x_j}$, is invertible then there exists $\epsilon > 0$ and a smooth curve $x_i = x_i(\lambda)$, $i=1, \dots, n$, which is a soln. of $f(x, \lambda) = 0$ in $|x| < \epsilon$, $|\lambda| < \epsilon$



For case in $\mathbb{R}^2$: $D_x f = 2 \times 2$ matrix



Fixed pts. fail to be hyperbolic on $\Delta=0$
and $\tau=0$ for $\Delta>0$

$\Delta=0$: steady state bifurcations

$\tau=0, \Delta>0$: Hopf bifurcations

Example of Hopf bifurcation that can be solved exactly:

$$\begin{aligned}\dot{x} &= \lambda x + \omega y - a(x^2 + y^2)x + b(x^2 + y^2)y \\ \dot{y} &= \lambda y - \omega x - a(x^2 + y^2)y - b(x^2 + y^2)x\end{aligned}$$

$x = y = 0$ is equilibrium for all λ

$$D_x f = \begin{bmatrix} \lambda & \omega \\ -\omega & \lambda \end{bmatrix}$$

$$\tau = 2\lambda$$

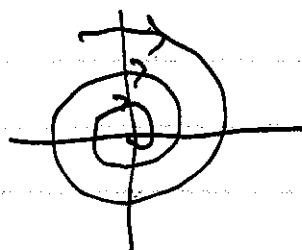
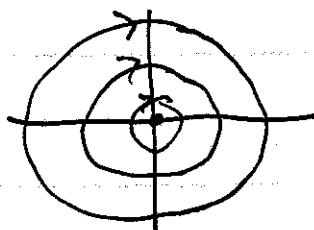
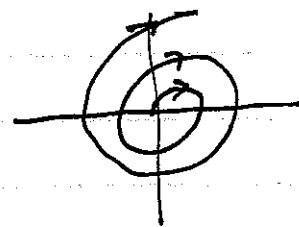
$$\Delta = \lambda^2 + \omega^2$$

eigenvalues: $\lambda \pm i\omega$

$\lambda < 0 \Rightarrow$ stable spiral

$\lambda = 0 \Rightarrow$ center (nonhyperbolic)

$\lambda > 0 \Rightarrow$ unstable spiral

Linear phase portraits: ($\omega > 0$) $\lambda < 0$  $\lambda = 0$ Hopf bifurcation
pt.can't conclude
anything from linear stability
calculation. $\lambda > 0$

Nonlinear problem: use polar coordinates

$$\left. \begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \end{aligned} \right\} \begin{aligned} r^2 &= x^2 + y^2 \\ \tan \theta &= y/x \end{aligned}$$

alternatively let $z = x + iy$, then $z = r \cos \theta + ir \sin \theta$
 $= r(\cos \theta + i \sin \theta)$
 $= r e^{i\theta}$

$$\dot{z} = \dot{x} + i\dot{y} = \dot{r} e^{i\theta} + ir\dot{\theta} e^{i\theta} = (\dot{r} + i\dot{\theta}r) e^{i\theta}$$

$$= \lambda(x + iy) + \omega(y - ix) - \underbrace{a(x^2 + y^2)}_{|z|^2}(x + iy) + b(x^2 + y^2)(y - ix)$$

$$x + iy = z = r e^{i\theta} \quad y - ix = -iz = -ir e^{i\theta}$$

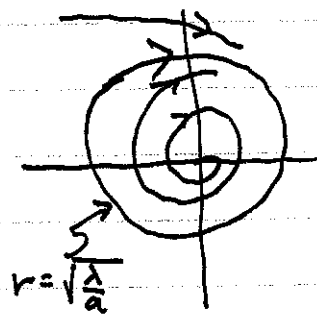
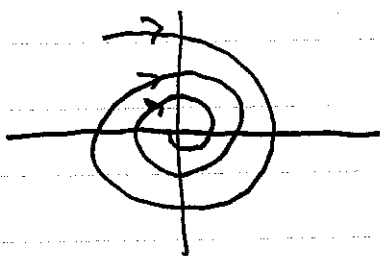
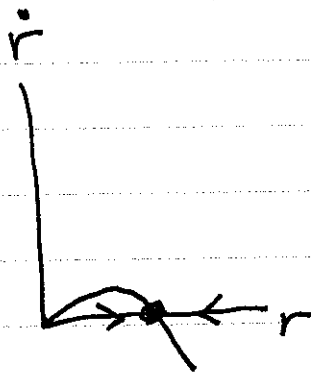
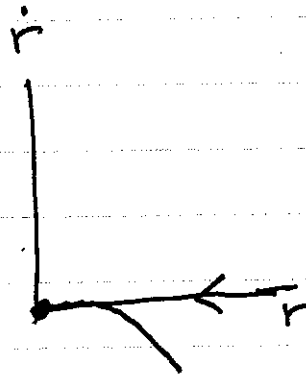
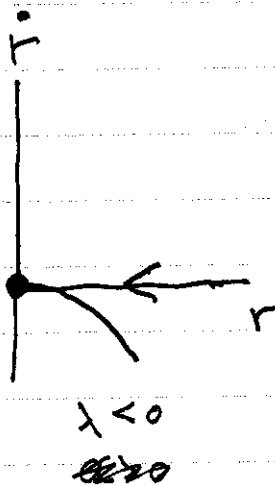
$$\dot{z} = (\lambda - i\omega)z - (a + ib)|z|^2 z$$

$$(\dot{r} + ir\dot{\theta})e^{i\theta} = (\lambda - i\omega)r e^{i\theta} - (a + ib)r^3 e^{i\theta}$$

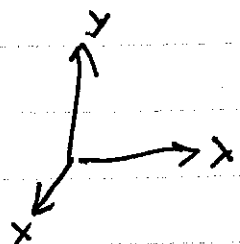
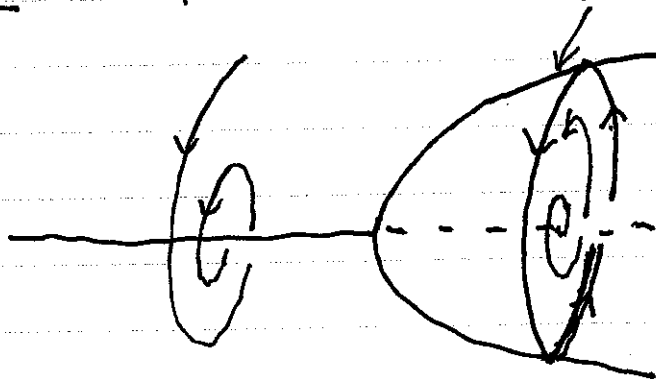
equating real & imaginary parts

$$\begin{aligned}\dot{r} &= \lambda r - ar^3 \\ \dot{\theta} &= -\omega - br^2\end{aligned}$$

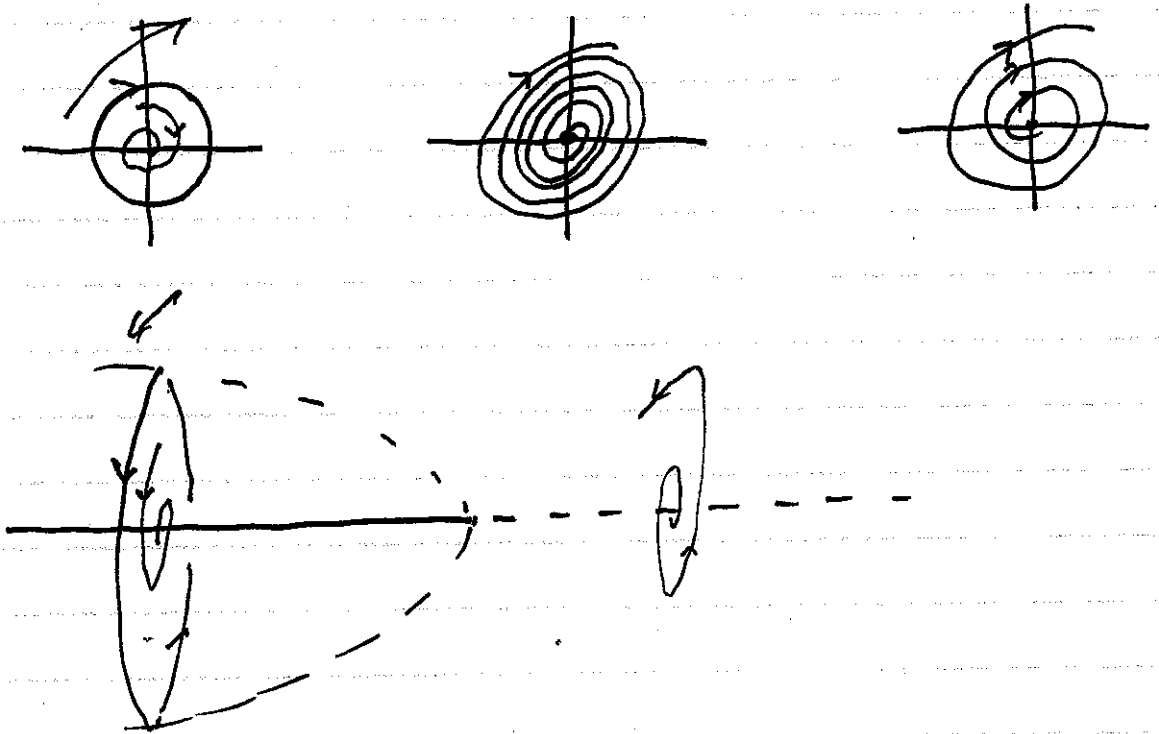
$a > 0$:



$a > 0$: supercritical Hopf bifurcation



$a < 0$: subcritical Hopf bifurcation



$a > 0$: supercritical , loss of stability of equilibrium associated with creation of a small amplitude stable limit cycle (amplitude r or $\sqrt{\lambda}$)

$a < 0$: subcritical , unstable limit cycle coexists with stable equilibrium & it vanishes at $\lambda = 0$